

Week 2: Solving Linear Equations

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Week 2 Quiz

2.1: One-variable inequalities; interval notation; number-line graphs

2.2: Compound inequalities (AND/OR); absolute value basics

2.3: Absolute-value equations; modeling simple contexts

Quick Reference: Solving Linear Equations

Steps for Solving Linear Equations

1. Simplify both sides (distribute, combine like terms)
2. Move variable terms to one side
3. Move constant terms to the other side
4. Divide or multiply to isolate the variable
5. Check your solution

Special Cases

- **One Solution:** When you get $x = \text{number}$
- **No Solution:** When you get a false statement (e.g., $5 = -3$)
- **Infinitely Many Solutions:** When you get a true statement (e.g., $0 = 0$)

Working with Fractions

To eliminate fractions, multiply both sides by the LCD (Least Common Denominator).

Week 2 Quiz: Solving Linear Equations

Part A: Solve (variables on one side)

Instructions: Solve for the variable:

1. $4(x + 3) = 28$

2. $0.25x - 7 = 6$

3. $\frac{x}{6} + 9 = 14$

Part B: Solve (variables on both sides)

Instructions: Solve for the variable:

4. $7x - 5 = 2x + 30$

5. $3(x - 2) = 2(x + 7)$

6. $5 - 2(x + 1) = x - 7$

Part C: Fractions & Special Cases

Instructions: Solve for the variable or identify special cases:

7. $\frac{x}{3} + 2 = \frac{x}{6} + 7$

8. $\frac{3}{4}x - 5 = \frac{1}{2}x + 1$

9. $2(x - 4) + 6 = 2(x - 4) - 3$

Part D: Word Problems

Instructions: Write and solve equations for the following problems:

10. A streaming service costs \$12 per month, plus a one-time \$6 activation fee. A different streaming service costs \$15 per month, with no activation fee. After how many months m will the total cost of both services be equal? Write and solve an equation to find the answer. (Include units in your final answer.)
11. A school fundraiser sells boxes of cookies for \$ x each. The club sold 18 boxes and raised a total of \$162. Write and solve an equation to find x , the price per box (in dollars).

Solution Guide

Part A: Solve (variables on one side)

1. $4(x + 3) = 28$

Solution:

Distribute the 4:

$$4x + 12 = 28$$

Subtract 12 from both sides:

$$4x = 16$$

Divide both sides by 4:

$$x = 4$$

Answer: $x = 4$

2. $0.25x - 7 = 6$

Solution:

Add 7 to both sides:

$$0.25x = 13$$

Divide both sides by 0.25:

$$x = 52$$

Answer: $x = 52$

3. $\frac{x}{6} + 9 = 14$

Solution:

Subtract 9 from both sides:

$$\frac{x}{6} = 5$$

Multiply both sides by 6:

$$x = 30$$

Answer: $x = 30$

Part B: Solve (variables on both sides)

4. $7x - 5 = 2x + 30$

Solution:

Subtract $2x$ from both sides:

$$5x - 5 = 30$$

Add 5 to both sides:

$$5x = 35$$

Divide both sides by 5:

$$x = 7$$

Answer: $x = 7$

5. $3(x - 2) = 2(x + 7)$

Solution:

Distribute on both sides:

$$3x - 6 = 2x + 14$$

Subtract $2x$ from both sides:

$$x - 6 = 14$$

Add 6 to both sides:

$$x = 20$$

Answer: $x = 20$

6. $5 - 2(x + 1) = x - 7$

Solution:

Distribute the -2 :

$$5 - 2x - 2 = x - 7$$

Combine like terms on the left:

$$3 - 2x = x - 7$$

Add $2x$ to both sides:

$$3 = 3x - 7$$

Add 7 to both sides:

$$10 = 3x$$

Divide both sides by 3:

$$x = \frac{10}{3}$$

Answer: $x = \frac{10}{3}$ or $x = 3\frac{1}{3}$
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Part C: Fractions & Special Cases

7. $\frac{x}{3} + 2 = \frac{x}{6} + 7$

Solution:

Multiply both sides by 6 (LCD of 3 and 6):

$$6 \cdot \frac{x}{3} + 6 \cdot 2 = 6 \cdot \frac{x}{6} + 6 \cdot 7$$

Simplify:

$$2x + 12 = x + 42$$

Subtract x from both sides:

$$x + 12 = 42$$

Subtract 12 from both sides:

$$x = 30$$

Answer: $x = 30$

8. $\frac{3}{4}x - 5 = \frac{1}{2}x + 1$

Solution:

Multiply both sides by 4 (LCD of 2 and 4):

$$4 \cdot \frac{3}{4}x - 4 \cdot 5 = 4 \cdot \frac{1}{2}x + 4 \cdot 1$$

Simplify:

$$3x - 20 = 2x + 4$$

Subtract $2x$ from both sides:

$$x - 20 = 4$$

Add 20 to both sides:

$$x = 24$$

Answer: $x = 24$

9. $2(x - 4) + 6 = 2(x - 4) - 3$

Solution:

Distribute on both sides:

$$2x - 8 + 6 = 2x - 8 - 3$$

Combine like terms:

$$2x - 2 = 2x - 11$$

Subtract $2x$ from both sides:

$$-2 = -11$$

This is a false statement, so there is no solution.

Answer: No solution

Part D: Word Problems

10. A streaming service costs \$12 per month, plus a one-time \$6 activation fee. A different streaming service costs \$15 per month, with no activation fee. After how many months m will the total cost of both services be equal?

Solution:

Set up the equation:

First service: $12m + 6$

Second service: $15m$

When costs are equal:

$$12m + 6 = 15m$$

Subtract $12m$ from both sides:

$$6 = 3m$$

Divide both sides by 3:

$$m = 2$$

Answer: $m = 2$ months

11. A school fundraiser sells boxes of cookies for \$ x each. The club sold 18 boxes and raised a total of \$162. Find x , the price per box.

Solution:

Set up the equation:

(number of boxes) $\times$ (price per box) = total raised

$$18x = 162$$

Divide both sides by 18:

$$x = 9$$

Answer: $x = \$9$ per box
